

RESEARCH ARTICLE

# Structural and Dynamic Determinants of BAK1 Mutational Sensitivity at the BH Motifs

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## Abstract

BAK1 (Bcl-2 homologous antagonist/killer) is a pro-apoptotic protein whose activation depends on conformational changes and dimerisation mediated by its conserved BH motifs. Missense mutations disrupting these regions have been linked to cancer, yet their structural consequences remain incompletely understood. Here, we investigated the mutational sensitivity of BAK1 by integrating pathogenicity prediction, structure selection, molecular docking, saturation mutagenesis, and ensemble-based conformational analysis. Per-position pathogenicity scores from AlphaMissense and DeMaSk converged on the BH1, BH2, and BH3 motifs as the most constrained regions of the sequence. Two experimental structures were selected for downstream analysis: 2IMT as a high-resolution free-protein reference, and 6UXR as a representative of the activated dimer. Molecular docking with SwissDock confirmed that 6UXR provides more favourable binding geometry for ligand interaction. Saturation mutagenesis with MutateX on both structures revealed that stability and binding sensitivity are not uniformly distributed across the BH motifs, with glycine residues and buried hydrophobic anchors showing the greatest intolerance to substitution. Critically, some variants showed decoupled effects, neutral for monomer stability but destabilising for the dimer interface, highlighting that single-state analyses can miss functionally relevant mutations. To account for conformational flexibility, an ensemble of eight structures derived from molecular dynamics simulations of PDB 2IMS was generated and used for additional MutateX calculations on the BH3 motif. The ensemble-averaged results confirmed the same principal hotspots identified in the single-structure analysis, with proline and bulky aromatic substitutions being systematically destabilising across conformations. Taken together, these results identify a small set of residues essential for both the structural integrity and dimerisation competence of BAK1, and show that combining multiple conformational states substantially improves the interpretation of missense variants in dynamic proteins.

**Keywords:** BAK1, apoptosis, BH motifs, saturation mutagenesis, conformational ensemble

## 1. Introduction

BAK1 is a multidomain pro-apoptotic effector protein and a key checkpoint in the intrinsic (mitochondrial) apoptotic pathway [1]. Under normal physiological conditions, BAK1 localises primarily to the mitochondrial outer membrane (MOM) and is maintained in an inactive conformation through interactions with anti-apoptotic BCL-2 family members (e.g. BCL-XL, MCL-1) and with VDAC2, which also serves to recruit newly synthesised BAK1 to the mitochondria [2, 3].

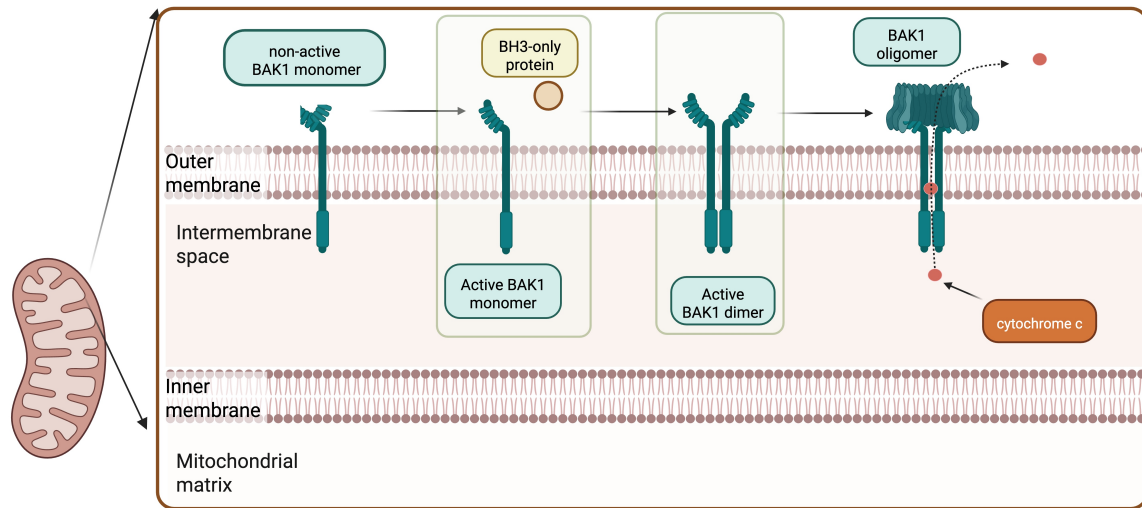
Upon receipt of cell death signals, BAK1 undergoes a series of conformational changes that lead to its oligomerisation and the formation of pores within the MOM. This process, known as mitochondrial outer membrane permeabilisation (MOMP), results in the release of apoptogenic factors into the cytosol, most notably cytochrome c, which activates caspase-9 and subsequently the downstream effector caspases, committing the cell to apoptosis [1, 4].

The molecular mechanism of BAK1 activation is described as a “hit-and-run” model (Figure 1) [5], which proceeds through four sequential steps. First, the BH3-only activator proteins, principally BID and BIM, dock into the BH3-binding groove of BAK1 in a tight but transient interaction. Second, this binding physically distorts BAK1’s conformation, exposing previously buried hydrophobic surfaces [5]. This causes the BAK1 monomer to change conformation, as the binding forces BAK1 to partially unfold: the N-terminus shifts position, BAK1’s own

BH3 domain becomes exposed, and a second interaction surface, located between helices  $\alpha 1$  and  $\alpha 6$ , opens up on the opposite face of the protein [1, 4]. Third, BID or BIM detaches from the groove (the “run”). The vacated BH3-binding groove is the same surface that the next BAK1 molecule will use to dock, meaning that by leaving, the activator frees the site for BAK1-to-BAK1 interactions; a single activator molecule can therefore sequentially activate multiple BAK1 monomers [5]. Finally, the activated BAK1 monomer uses its newly exposed BH3 domain to insert into the groove of another activated BAK1 molecule, forming a symmetric dimer. Dimers then assemble into higher-order oligomers, ultimately forming a pore large enough for cytochrome c to pass through [1, 4].

### Structural Organisation

The first crystal structure of BAK1 (PDB 2IMS/2IMT) was solved by Moldoveanu et al. in 2006, revealing that the protein adopts a globular fold composed of nine  $\alpha$ -helices, with a central hydrophobic core helix ( $\alpha 5$ ) surrounded by amphipathic helices and a C-terminal transmembrane helix that anchors the protein to the MOM [6]. This structure also provided the framework for assigning the three BCL-2 homology (BH) motifs that define BAK1 as a pro-apoptotic effector: BH1 (residues 117–136), BH2 (residues 169–184), and BH3 (residues 74–88), as catalogued in UniProt [7]. These motifs were originally identified through sequence alignments across the BCL-2 family and shown to be functionally critical for protein–protein interactions within



**Figure 1.** Structural progression of BAK1 activation during apoptosis. Image created in Biorender.

the family [1]. The BH domains mediate both homo- and heterodimerisation with other BCL-2 family members, and an intact BH3 motif in particular is strictly required both for BAK1's pro-apoptotic activity and for its interactions with anti-apoptotic partners [1].

BAK1 is expressed across a wide variety of tissues, with the highest expression levels reported in the heart and skeletal muscle [8].

#### Disease Associations

BAK1 dysregulation is most strongly associated with cancer, where loss of its pro-apoptotic function allows tumour cells to evade programmed cell death and contributes to chemotherapy resistance [1]. Variants in BAK1 have been catalogued across several cancer-focused databases, including ClinVar [9], cBioPortal [10], and COSMIC [11], which together yielded 121 cancer-associated variants compiled through the MAVISp framework [12] and used as the starting point for the analyses presented in this study.

## 2. Materials and Methods

### 2.1. Prediction of pathogenicity

To assess the sensitivity of each residue to missense substitution, per-position pathogenicity was computed using two complementary predictors: **AlphaMissense** [13], a structure- and evolution-aware deep learning model, and **DeMaSk** [14], an alignment-based method that infers fitness effects from homologous sequences. For each residue, the centered rank-normalized damaging score was averaged over all 19 possible substitutions, and the standard deviation across substitutions at that position was retained as a measure of variability. Positive values indicate residues at which mutations are predicted to be damaging on average, while negative values mark mutation-tolerant positions.

### 2.2. Variant selection

To identify relevant mutations for the BAK1 protein, a multi-database consensus approach was employed. The study's scope was restricted to the **BH1, BH2, and BH3 motifs**, identified as the most mutationally sensitive regions in Section 3.1. While a full mutational saturation scan of these motifs would represent the most exhaustive approach, its significant computational burden necessitates a prioritized selection of variants based on real-life clinical samples. Variants were thus retrieved from the ClinVar, COSMIC, and cBioPortal databases [9–11]. The following successive filters were applied to refine the selection:

- Only Single Nucleotide Variants (SNVs) classified as missense variants were considered.

- The scope was restricted to residues located within the conserved BH motifs of BAK1.
- Only variants identified in at least two of the three consulted databases were retained to ensure a robust selection.

This methodology yielded a final set of 27 variants distributed across 22 residues (see Table 1 in Appendix A).

While the 27 clinical variants formed the priority subset for initial validation, the analysis was ultimately extended to a complementary subset comprising all other possible substitutions within the selected domains. This approach ensured a full mutational saturation scan, providing a comprehensive landscape of the BH motifs' energetic and structural tolerance.

### 2.3. Structure Selection

The available BAK1 structures for UniProt ID Q16611 were screened using PDBMiner to identify models suitable for downstream analysis. Structures were filtered according to experimental method, resolution, R-free, mutation status, whether they represented a free protein or a complex, and their coverage of the regions of interest. X-ray diffraction structures were prioritized because they provide higher local confidence and clearer atomic detail than lower-resolution or purely predicted models. **AlphaFold** [15] and **AlphaFold3** [16] were then used as comparison models to assess agreement with the experimental structures.

To assess the ligand-binding compatibility of the selected BAK1 structures, molecular docking was performed using SwissDock [17,18]. The **2IMT** and **6UXR** structures were prepared as clean receptor models by removing crystallization ligands, and docking was carried out using a small-molecule probe ligand to compare binding-site accessibility between the two conformations. The resulting interactions were evaluated using calculated affinity values (kcal/mol), where more negative values indicate more favourable binding.

### 2.4. Effect of mutations on stability

The impact of point mutations on BAK1 stability was estimated using the **MutateX** automated pipeline [19], built around **FoldX** (suite5) [20,21]. The analysis was performed on the **2IMT** structure selected in Section 2.3, after removal of all solvent molecules. Upon activation of the required conda environment, the MutateX pipeline was executed within a Python 3.12.9 environment with parallelisation on 4 threads and *deep* cleaning mode. MutateX performs *in silico* saturation mutagenesis, computing the change in folding free energy ( $\Delta\Delta G$ ) for every possible amino acid substitution at each selected position.

Following the per-position pathogenicity analysis (Section 2.1) and the gene and variant selection (Section 2.2), the analysis was restricted to the residues spanning the BH1 and BH3 motifs of BAK1. As for the local interaction analysis, the pipeline was executed in two stages: an initial run on the priority subset of variants, followed by a complementary run to achieve full mutational saturation of the targeted motifs. At each position, all 20 amino acids were modelled (including the wild-type residue, which serves as an internal control with  $\Delta\Delta G \approx 0$ ), and per-mutation  $\Delta\Delta G$  values were averaged over five independent FoldX runs to reduce stochastic variability.

The energetic data were then processed using the `ddg2` visualisation tools, notably `ddg2excel` and `ddg2heatmap`. The averaged  $\Delta\Delta G$  values were visualised as a per-residue heatmap across the selected positions, providing a residue-level view of stability sensitivity to substitution. To ensure data interpretability and focus on biologically relevant hotspots, the heatmap energy scale was restricted to the range  $-2$  to  $+5$  kcal/mol.

## 2.5. Effect of mutations on local interaction using MutateX

Following the same computational framework described in the previous section, we extended the MutateX analysis to evaluate the local interactions of the BAK1 activated dimer. This study was performed using the alternative dimeric structure 6UXR (Section 2.3) as the primary input after the removal of all solvent molecules.

Upon activation of the required conda environment, the MutateX pipeline was executed within a Python 3.12.9 environment using FoldX (suite5). While the core algorithm remained the same as in the previous section (parallelization on 4 threads and *deep* cleaning mode), the pipeline was specifically configured for interface interaction by incorporating the `-B` argument for binding energy calculations and the `-I` option to provide the required interface template file.

The analysis was restricted to a subset of 50 residues (positions 69-88 and 117-146 on chain A). This selection covers the BH3 and BH1 regions of interest, but was extended to include the full length of the corresponding  $\alpha$ -helices resolved in the 6UXR structure. These interface residues were also confirmed based on their proximity to the partner subunit in the dimer (6UXR), as described in the Figure 6C. Saturation mutagenesis was then performed for all 19 alternative amino acids at each position, and the resulting  $\Delta\Delta G_{binding}$  values were averaged over multiple FoldX runs.

Finally, the energetic data were processed using the `ddg2` visualisation tools, notably `ddg2excel` and `ddg2heatmap`. To ensure data interpretability and focus on biologically relevant hotspots, thresholds of  $+5$  kcal/mol and  $-2$  kcal/mol were applied to the heatmap energy scale.

Following the selection method described in Section 2.2, the pipeline was executed in two stages: an initial run on the priority subset, followed by a complementary run to achieve full mutational saturation of the targeted motifs.

## 2.6. Molecular dynamics and ensemble generation

To capture the conformational dynamics of the BAK1 protein, molecular dynamics (MD) trajectories were retrieved from the ATLAS database. The selected simulation data corresponded to the high-resolution X-ray structure of BAK1 chain A (PDB ID: 2IMS). Prior to the MD simulations, structural refinement was performed using **MODELER** to reconstruct short gaps of missing residues. The overall stability and structural integrity of the simulated protein were validated by analyzing standard trajectory metrics, including the Root Mean Square Deviation (RMSD), radius of gyration, residue-residue contact maps, and Ramachandran plots.

For the generation of a representative conformational ensemble, trajectory processing and clustering were performed using the **GROMACS** software suite. To ensure a robust and comprehensive sampling of the conformational landscape, three independent MD replicates, initiated with different starting seeds, were concatenated into a single continuous trajectory. Subsequently, an all-to-all RMSD matrix was computed. Finally, conformational clustering was executed employing the GROMOS algorithm. An RMSD distance cutoff of 0.24 nm, derived from the average RMSD of the combined trajectory, was applied to partition the frames and extract the representative cluster centers used for downstream ensemble-mode mutational analyses.

## 2.7. Ensemble-mode stability and local interaction

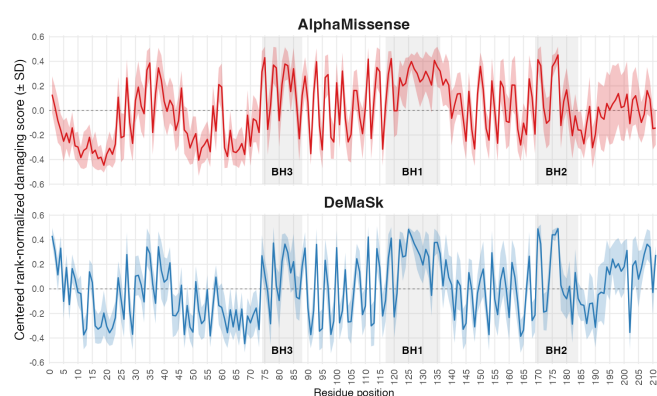
In order to assess the effects of mutations while also taking into account the protein stability, a series of MutateX calculations were performed in the ensemble mode using the eight clusters structure obtained from the molecular dynamics clustering analysis. Saturation mutagenesis was carried out on the BH3 motif for each cluster structure using the MutateX pipeline together with FoldX (suite5). The resulting  $\Delta\Delta G$  values were averaged across the ensemble structures. With a single-structure approach, this method allows mutation effects to be evaluated over multiple conformations sampled during the simulation, giving a more realistic estimation of mutational stability effects.

# 3. Results

## 3.1. Pathogenicity landscape highlights the BH motifs

Per-position pathogenicity profiles from AlphaMissense and DeMaSk show strong qualitative agreement along the BAK1 sequence despite their methodological differences (Figure 2), lending confidence to the inferred pathogenicity landscape. Both methods identify the three Bcl-2 homology motifs (BH3, BH1, and BH2; shaded in grey) as regions of elevated damaging score, consistent with their established roles in mediating heterodimerization and pore formation. The BH1 region shows the most pronounced peak in both predictors, with mean scores reaching  $\sim 0.2$ – $0.4$ , in line with the high evolutionary conservation of its core hydrophobic residues. A further stretch of elevated pathogenicity is observed around residues 30–45, where both predictors show concordant scores well above the protein-wide mean, suggesting additional functionally relevant positions outside the canonical BH motifs. Towards the C-terminus, an elevated signal is also observed in the membrane-anchoring transmembrane region, particularly in DeMaSk, consistent with the strong evolutionary conservation expected for residues that mediate stable insertion into the mitochondrial outer membrane.

A striking feature of the pathogenicity profile within the BH motifs and, to a lesser extent, across other structured stretches, is that the damaging score oscillates with a short periodicity rather than remaining uniformly elevated. Peaks of high predicted pathogenicity alternate with adjacent positions of much lower score, producing a regular zig-zag pattern. This periodicity is characteristic of  $\alpha$ -helical secondary structure: with  $\sim 3.6$  residues per helical turn, consecutive residues face alternately inward (toward the hydrophobic core or a binding interface) and outward (toward the solvent). The inward-facing residues, which appear as peaks in the profile, are under strong selective pressure because they contribute directly to helix packing, hydrophobic core stability, and the protein-protein contacts that the BH motifs mediate. Substitutions at these positions are therefore predicted as damaging by both methods. The outward-facing residues, by contrast, are solvent-exposed, make few structural or functional contacts, and consequently tolerate substitution with little predicted impact, giving rise to the troughs of the oscillation. The fact that this helical signature is recovered independently by a structure-aware predictor (AlphaMissense) and a purely evolution-based one (DeMaSk) indicates that it



**Figure 2.** Per-position pathogenicity scores for BAK1 from AlphaMissense and DeMaSk. Shaded bands indicate the standard deviation across all 19 possible substitutions at each position; grey shaded regions mark the BH3, BH1, and BH2 motifs.

reflects a genuine biophysical and evolutionary constraint rather than an artefact of either method, and it provides residue-level support for the  $\alpha$ -helical nature of the BH1, BH2, and BH3 segments.

Overall, the  $\alpha$ -helical zig-zag signature is present throughout the structured regions of the protein. What distinguishes the BH motifs is not the presence or absence of this pattern itself, but the consistently elevated mean pathogenicity sustained across a contiguous stretch of residues, indicating that the residues in these motifs contribute more to the protein's function/stability than their counterparts in the inter-domain linkers. This sustained elevation, rather than the oscillation alone, is what marks the BH regions as the most mutationally sensitive parts of BAK1, and it guided the selection of the motifs as the focus of the subsequent stability and binding analyses.

## 3.2. Structure selection

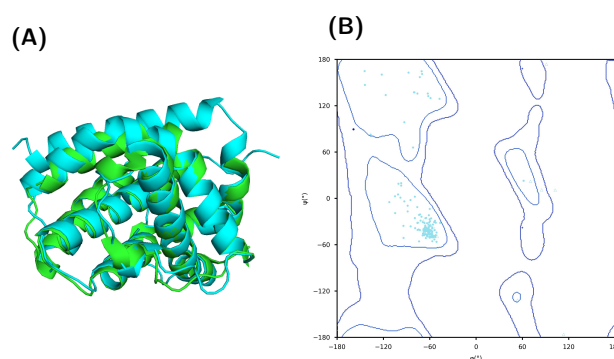
### 3.2.1. Free Protein Structure Selection

Among the available structures, 2IMT was selected as the main structure for downstream analysis. It is an X-ray structure with 1.49 Å resolution and an R-free value of 0.2201, and it contains no reported mutations. Chain A covers residues 21–183, giving good coverage of the protein region of interest and making it the best free-protein reference structure for the analysis.

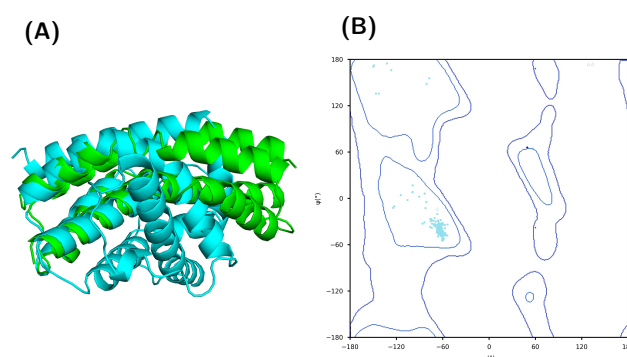
The Ramachandran plot for 2IMT showed that most residues fall within favoured regions, indicating good stereochemical quality Figure 3(B). The AlphaFold comparison showed strong overlap with the experimental structure, supporting the conclusion that 2IMT represents a biologically realistic conformation Figure 3(A). These results justify its use as the primary structure for the main part of the analysis.

### 3.2.2. Supporting Complex Structure

The Ramachandran plot for 6UXR (Figure 4b) also indicated good overall quality, although it was slightly less complete than 2IMT. The AlphaFold comparison (Figure 4a) showed a good core match but some helix shifts near the lipid region, consistent with a different conformational state. This was expected, as the 6UXR represents the active homodimeric conformation of the BAK1 protein. For this reason, 6UXR was used as a supporting structure rather than as the main reference.



**Figure 3.** Structural quality assessment for 2IMT; (A) Comparison of a single chain of 2IMT structure (green) to the AlphaFold structure of BAK1 (blue), (B) Ramachandran plot of the 2IMT BAK1 structure.



**Figure 4.** Structural quality assessment for 6UXR; (A) Comparison of a single chain of 6UXR structure (green) to the AlphaFold structure of BAK1 (blue), (B) Ramachandran plot of the 6UXR BAK1 structure.

### 3.2.3. Final structure selection

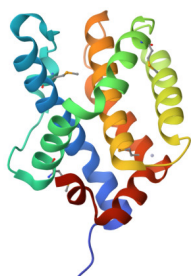
Overall, 2IMT was selected as the primary structure because it provided the best combination of resolution, R-free, lack of mutations, and coverage for analysis of the free protein form. 6UXR was included because it adds value for local interaction and dimer-related analysis. Using both structures allows the study to compare the free protein state with a more interaction-relevant state.

### 3.2.4. SwissDock Affinity Analysis

The SwissDock results suggested stronger ligand-binding compatibility for 6UXR than for 2IMT, with 6UXR giving consistently negative affinity values down to -4.561 kcal/mol, whereas 2IMT showed positive values. This supports the use of 6UXR for local interaction analysis, while 2IMT remains the better reference for the free-protein state.

### 3.2.5. Ensemble-generation structure

Finally, 2IMS was included as the starting structure for ensemble generation because it was the only ATLAS hit for Bak1 and represents an activated BAK1 conformation. Its PDB quality metrics were checked to confirm that it was suitable for the ensemble analysis.



**Figure 5.** Structure 2IMS of the *Bak1* protein, chain A

### 3.3. Effects of mutations on stability

The range of residues investigated for stability effects therefore comprised the BH motifs of BAK1, whose functional relevance was already established in the literature and further confirmed by the pathogenicity analysis presented in Section 2. The analysis was performed on the 2IMT structure, selected in Section 3.2.1. Within these regions, side-chain solvent accessibility was used to refine the focus of the analysis: while surface-exposed residues are typically involved in mediating intermolecular interactions, the buried and intermediate residues (Figure 6A) are of greater interest from a structural standpoint, as they contribute directly to helix packing and hydrophobic core stability and are therefore the positions where substitutions are most likely to compromise fold integrity.

#### 3.3.1. Selection of residues for stability analysis

To enable a consistent comparison between functional states, the stability analysis was restricted to residues present in both selected structures (PDB IDs 2IMT and 6UXR), corresponding to the overlapping region spanning positions 69–148. Within this range, the focus was placed on the BH motifs of BAK1, whose functional relevance was already established in the literature and further confirmed by the pathogenicity analysis presented in Section 3.1. Side-chain solvent accessibility was then used to refine the selection: while surface-exposed residues are typically involved in mediating intermolecular interactions, the buried and intermediate residues (Figure 6A) are of greater interest from a structural standpoint, as they contribute directly to helix packing and hydrophobic core stability and are therefore the positions where substitutions are most likely to compromise fold integrity. The final set of residues selected for the stability analysis is highlighted on the 6UXR BAK1 structure in Figure 6C.

Beyond the choice of positions themselves, interpreting the impact of substitutions requires close attention to the chemical nature of the change, that is, both the wild-type residue being replaced and the residue being introduced. The destabilising effect of a substitution depends not only on where it occurs in the structure, but also on the physicochemical compatibility between the new side chain and its local environment: features such as size, hydrophobicity, charge, aromaticity, and conformational rigidity all influence whether a substitution can be accommodated without disrupting helix packing, hydrogen-bonding networks, or the hydrophobic core. The mutational scan therefore captures a matrix of context-dependent effects, in which both the identity of the original residue and that of its replacement determine the predicted change in stability.

Mutations were classified according to their predicted effect on stability based on the computed  $\Delta\Delta G$  values: stabilising ( $\Delta\Delta G \leq -3$  kcal/mol), neutral ( $-2$  kcal/mol  $\leq \Delta\Delta G < 2$  kcal/mol), uncertain ( $2$  kcal/mol  $\leq \Delta\Delta G < 3$  kcal/mol or  $-3$  kcal/mol  $\leq \Delta\Delta G < -2$  kcal/mol), and destabilising ( $\Delta\Delta G \geq 3$  kcal/mol). For visualisation, the heatmap colour scale was restricted to the range  $-2$  to

$5$  kcal/mol, focusing the analysis on neutral, uncertain, and destabilising substitutions, stabilising mutations were therefore not examined in detail.

#### 3.3.2. Per-residue stability landscape

The per-residue stability heatmap obtained from the MutateX scan is shown in Figure 6B, with each column representing one of the analysed positions and each row a possible amino acid substitution. Several broad patterns emerge from the distribution of  $\Delta\Delta G$  values across the heatmap, reflecting the principles outlined above.

The per-residue stability heatmap reveals landscapes for the two motifs. The BH3 motif is broadly tolerant to substitution at most central positions (LA78, AA79, IA80, IA81), with only specific substitutions, primarily bulky aromatics (F, W, Y) and proline, predicted as severely destabilising due to steric clashes and disruption of  $\alpha$ -helical geometry. In contrast, GA75 shows general intolerance to substitution, with virtually all 19 alternative amino acids predicted as severely compromising fold stability, suggesting that the absence of a side chain at this position is structurally essential.

The BH1 motif displays a markedly different pattern, with most positions predominantly destabilising across the substitution range. Hydrophobic anchor residues (LA118, FA119, IA123, FA134, YA136) tolerate only conservative replacements with similarly bulky hydrophobic side chains, consistent with their role in helix packing and hydrophobic core stability. WA125 illustrates this constraint particularly well: only substitutions to other bulky hydrophobic residues (M, F) are predicted as neutral, reflecting the requirement for a large non-polar side chain to fill its buried position in the hydrophobic core. A notable exception within BH1 is GA126, which behaves as an internal control due to its solvent-exposed orientation: most replacements are predicted as neutral or near-neutral, consistent with the absence of strong packing constraints at this position.

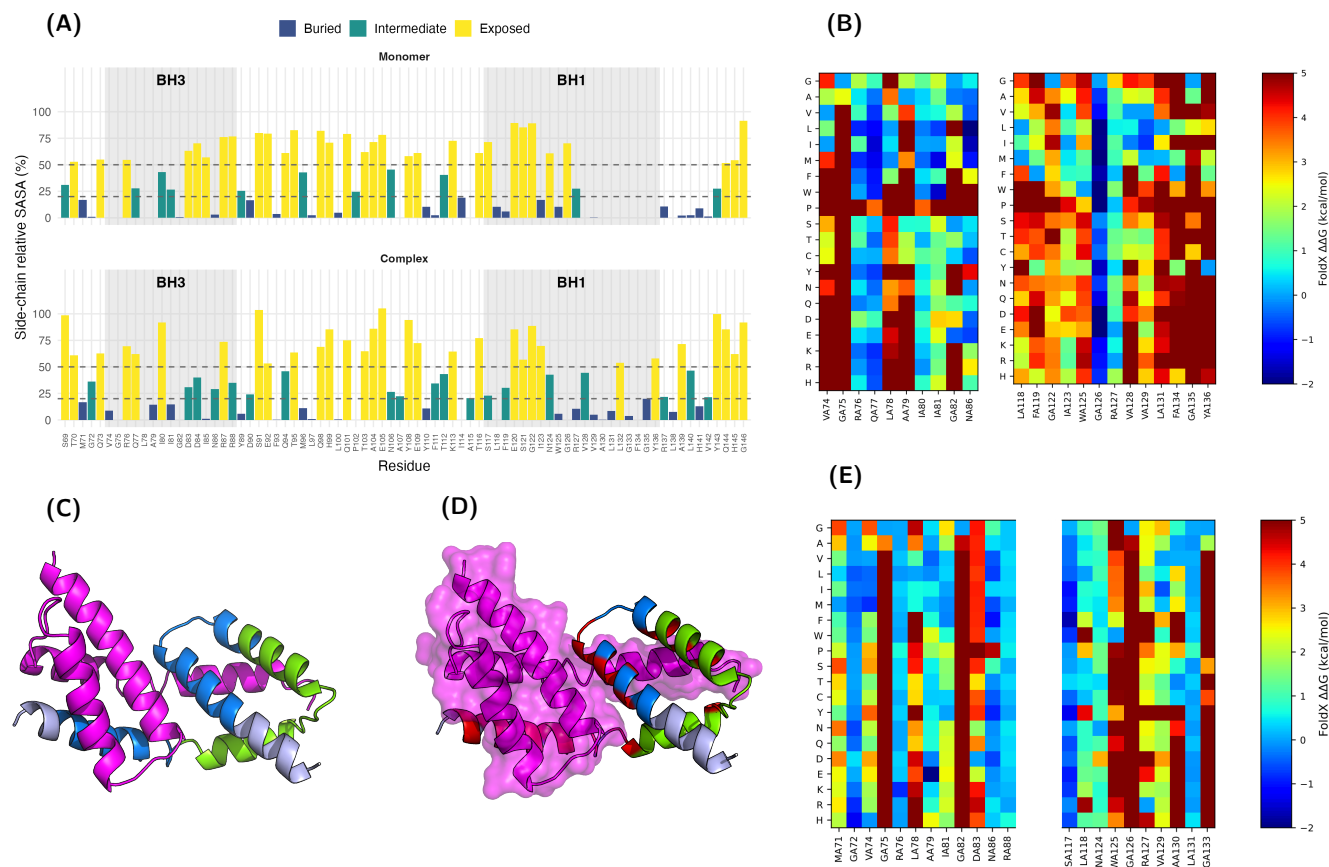
Across both motifs, two general trends emerge. First, proline substitutions are consistently destabilising at every position, reflecting the incompatibility of proline's cyclic backbone with  $\alpha$ -helical geometry. Second, the introduction of bulky aromatic side chains (F, W, Y) at positions occupied by smaller wild-type residues, and conversely the replacement of bulky aromatics with smaller residues, both produce strong destabilisation. This highlights the importance of side-chain compatibility for maintaining the structural integrity of the BH motifs.

### 3.4. Effects of mutations on local interactions

#### 3.4.1. Interface characterization and residue selection

To evaluate the impact of substitutions on the functional form of BAK1, we employed the activated dimeric structure 6UXR. In this conformation, the interaction is primarily mediated by the reciprocal docking of the BH3 domain from one subunit into the previously buried, hydrophobic BH1-binding groove of the partner monomer (Figure 6C). It was therefore logical to investigate mutations within these specific motifs to determine their impact on local interactions. Initially, flanking residues (69–73 and 137–146) were also included because they belong to the same  $\alpha$ -helices as the BH1 and BH3 domains. This allowed us to easily incorporate the terminal residues resolved in the 6UXR structure, ensuring that the helical extremities were fully represented.

However, the selection of residues for the binding analysis was also refined by identifying the physical contact surface. As shown in Figure 6D, the interface was defined as residues on Chain A within a  $5 \text{ \AA}$  radius of the partner subunit. This structural filtering confirmed that the interaction core resides within the BH1 and BH3 domains. Notably, while the N-terminal extension of the BH3 motif (residues 69–73) was included due to its direct involvement in the dimer interface, the C-terminal extension of the BH1 motif (residues 137–146) was excluded



**Figure 6.** Composite analysis of BAK1 stability and structural features. **(A)** SASA distribution across residues. **(B)** MutateX  $\Delta\Delta G_{\text{stability}}$  heatmap (folding stability). **(C)** Activated BAK1 dimer (PDB: 6UXR). BH1 and BH3 motifs are highlighted in navy blue, with extended study regions in pastel blue. **(D)** Dimer interface visualization. Chain B is shown as a semi-transparent surface; residues on Chain A within 5 Å of the partner subunit are highlighted in red. **(E)** MutateX  $\Delta\Delta G_{\text{bind}}$  heatmap (local interface interaction).

from the binding calculations, as these positions do not contribute to the identified contact surface. As expected, residues showing high side-chain solvent accessibility in the complex (Figure 5A) were also excluded from the interaction analysis, as their exposure indicates they are not involved in the buried subunit-subunit interface.

Following the MAVISp framework guidelines, the energetic impact on binding was categorized using a conservative threshold of 1 kcal/mol. Substitutions were classified as stabilizing for the interface if  $\Delta\Delta G_{\text{bind}} < -1$  kcal/mol and destabilizing if  $\Delta\Delta G_{\text{bind}} > 1$  kcal/mol. Values falling within the  $[-1, 1]$  range were considered neutral, representing mutations that do not significantly alter the binding affinity of the dimer.

### 3.4.2. Binding energy landscape and hotspot identification

The binding energy landscape (Figure 6E) reveals the sensitivity of the BAK1 dimer interface, with the BH3 motif (left) and BH1 motif (right) separated by a structural gap. Overall, the heatmap shows a high density of destabilizing mutations ( $\Delta\Delta G_{\text{bind}} > 1$  kcal/mol), particularly at conserved positions where side-chain substitution appears strictly prohibited for interface complementarity.

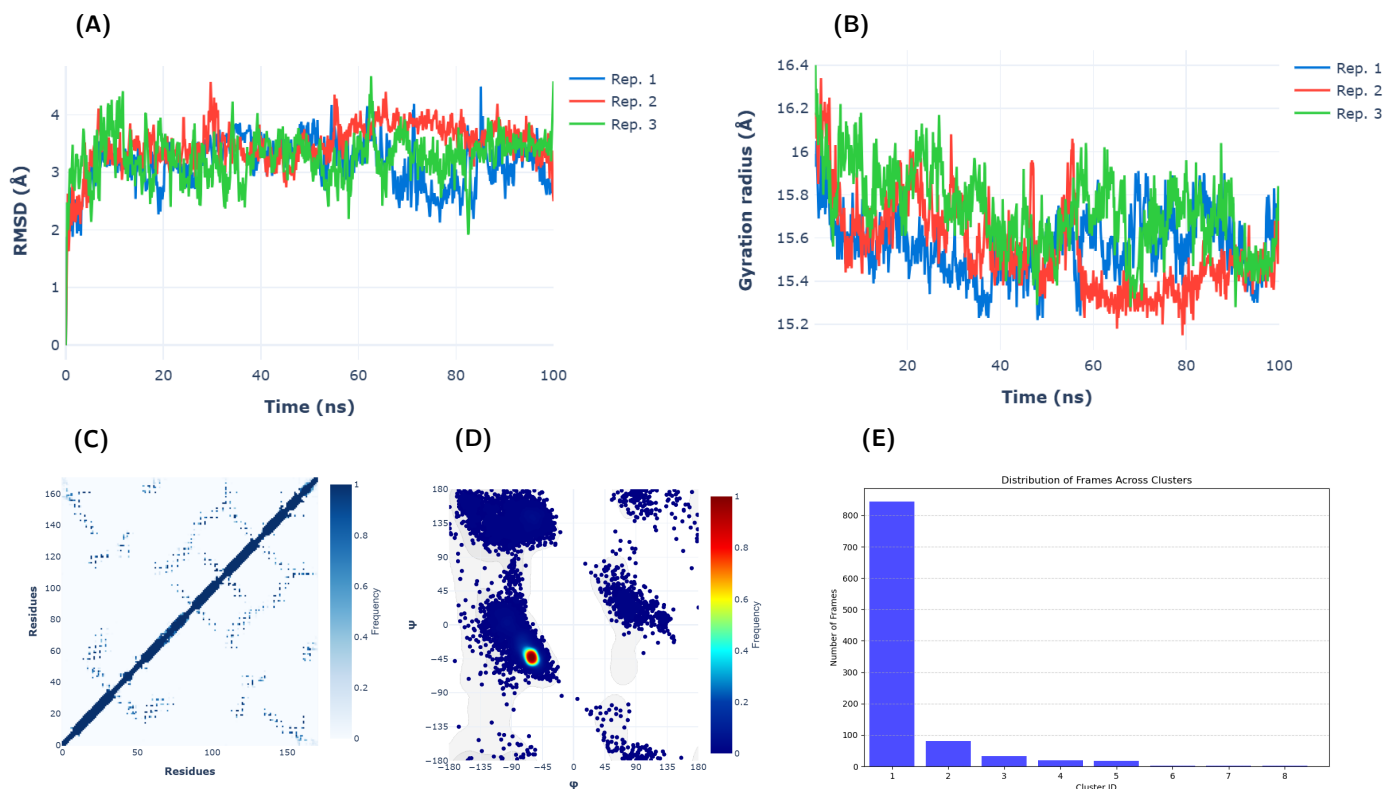
A striking feature of this landscape is the systematic intolerance at glycine positions (GA75, GA82, GA126, and GA133). They appear as solid dark-red columns, indicating that virtually any substitution severely compromises the binding affinity, regardless of the chemical nature of the new amino acid. As Glycine is the smallest amino acid, these residues are likely buried in high-density packing zones, where

the absence of a side chain is a non-negotiable structural requirement to avoid steric hindrance between the two monomers. Interestingly, GA72 shows a remarkable lack of sensitivity to all mutations despite being located at the dimer interface. A possible explanation is that this residue most likely occupies a more peripheral or flexible region of the interface where steric constraints are less strict, accommodating most side chains without loss of binding.

Within the BH3 domain and its extensions, residues display varying degrees of structural necessity. MA71, despite being located outside the core BH1 motif, shows high sensitivity to most substitutions, confirming that the N-terminal flanking regions are essential for proper interface docking. In the motif core, LA78 represents a selective profile: it is highly destabilized by polar or charged residues but maintains some plasticity toward conservative hydrophobic replacements. In contrast, DA83 acts as a near-absolute anchor; almost all substitutions result in significant binding loss, though the  $\Delta\Delta G$  magnitudes remain slightly below the extreme levels observed for glycines.

The BH1-binding groove exhibits a similar hierarchical organization. AA130 follows a strict volume-dependent pattern, where the majority of mutations are highly destabilizing, yet the site remains permissive to a narrow range of residues with small side chains. Conversely, WA125 emerges as a critical hotspot with total intolerance across the amino acid spectrum. This residue likely functions as a primary hydrophobic "plug", where any change in volume or chemistry disrupts the lock-and-key fit with the BH3 helix.

These identified hotspots demonstrate that while the interface allows for some local flexibility, specific non-glycine anchors are also mandatory for dimer stability. In contrast, residues such as RA88 and SA117 exhibit high mutational plasticity, which correlates with their exposed status in the complex (Figure 5A). RA127, despite being categorized



**Figure 7.** Molecular Dynamics (MD) simulation metrics. **(A)** RMSD versus time across three replicates. Variations remain within  $\sim 2$  Å, indicating the protein remains folded throughout the simulation. **(B)** Radius of gyration versus time. Fluctuations of  $\sim 1$  Å confirm the protein remains compact. **(C)** Residue-residue contact map averaged over the simulation. **(D)** Ramachandran plot of backbone dihedral angles sampled during the simulation. **(E)** Distribution of frames across conformational clusters. The protein occupies Cluster 1 for the vast majority of the simulation, with brief transient excursions to minor clusters, further confirming high structural stability.

as intermediate, also appears permissive, suggesting these positions occupy peripheral or non-critical regions where substitutions do not significantly impact the binding affinity.

## 3.5. Ensemble generation

### 3.5.1. Structure choice

For the ensemble generation, the first step was to search the selected protein in the **ATLAS** database. To do so, we used the name of the protein *Bak1* and only one result was available: the chain A of the **2IMS** structure from **PDB** (Figure 5). As this is a different structure from the one selected in the previous parts, we checked in the **RSCB PDB** database the metrics of this structure to see to what extent the result will be representative. Indeed, if the structure has poor metrics, meaning the protein is poorly represented, then the ensemble generation and analysis won't be relevant.

The PDB structure **2IMS** was deposited in 2006 and corresponds to chain A of the Bak1 protein, covering a sequence of 171 residues. The structure was determined using X-ray diffraction and has a high resolution of 1.48 Å, which is well below the commonly acceptable threshold of 2.5 Å for detailed structural analysis. The reported R-value for free reflections is 0.204, while the R-value for the working set is 0.179, and the observed R-value is 0.180. The small difference between R-free and R-work indicates a reliable fit of the model to the experimental data. Overall, these metrics suggest that the structure is of sufficient quality to perform meaningful ensemble and molecular dynamics analyses. It is also important to note that this monomer originates from the active dimer form of the Bak1 protein.

### 3.5.2. Molecular Dynamics analyses

Before proceeding with the Ensemble Generation, we evaluated the molecular dynamics of the structure using metrics from the **ATLAS** database (Figure 7). It is important to note that prior to the MD simulation itself, a structural refinement step was performed using **MODELER** to reconstruct missing residues. This software is specifically suited for modeling short gaps ("no more than 5 consecutive gaps or modified residues" [22]).

The first metrics we'll take a look at are the Root Mean Square Distance (RMSD) (Figure 7a) and the Gyration Radius (Figure 7b) evolutions during the simulation. We can observe that both metrics are mostly constant during the simulation, with RMSD in the interval  $[2\text{Å}, 4\text{Å}]$  and the gyration radius in the interval  $[15.2\text{Å}, 16.4\text{Å}]$ . Such small variations are unlikely due to major unfolding, thus the 3D structure seems to be maintained during the simulations.

Then, we can look at the contact map (Figure 7c). We can see that, apart from the diagonal, all the points that we can see have a frequency  $> 0.8$ , i.e. a probability higher than 80% for these parts to stay locked together. Thus this contact map corroborate the idea that the protein isn't going under major unfolding events during the MD simulation.

Finally, the Ramachandran plot from Figure 7d emphasize that. We can observe that the "hot spot", i.e. the points of high frequency, are extremely localized and tightly clustered in favored regions of a standard Ramachandran plot. Hence, the secondary structures of the proteins seem to stay locked during the simulation, corroborating that the protein overall stay folded during the MD simulation.

Following the quality assessment of the MD simulations, ensemble generation was performed using **GROMACS** through a three-step procedure.

First, to ensure a comprehensive sampling of the conformational space, the three independent replicates (generated using different starting seeds) were concatenated into a single continuous trajectory using the `gmx trjcat` command.

Next, an RMSD matrix was computed for this combined trajectory using `gmx_mpi rms`. The protein main chain was selected for both the least-squares structural alignment and the subsequent RMSD calculation. This analysis yielded a minimum RMSD of 0, a maximum of 0.526897, and an average of 0.240485.

Finally, conformational clustering was executed via the `gmx_mpi cluster` command, employing the calculated average RMSD as the cutoff threshold.

From this clustering, 8 conformations are highlighted. However, it seems that one conformation is highly favored as the protein spent more than 85% of the time in it during the simulations. It means that the protein is very stable and that it doesn't unfold. However, we can see in the heatmap (Figure 8) that from time to time, very occasionally, the protein is going to go from the main conformation to another the time of an instant. But, during the vast majority of the time, it'll remain in the cluster 1 conformation, which is corroborating the histogram plot.

### 3.6. Ensemble-mode mutational stability analysis

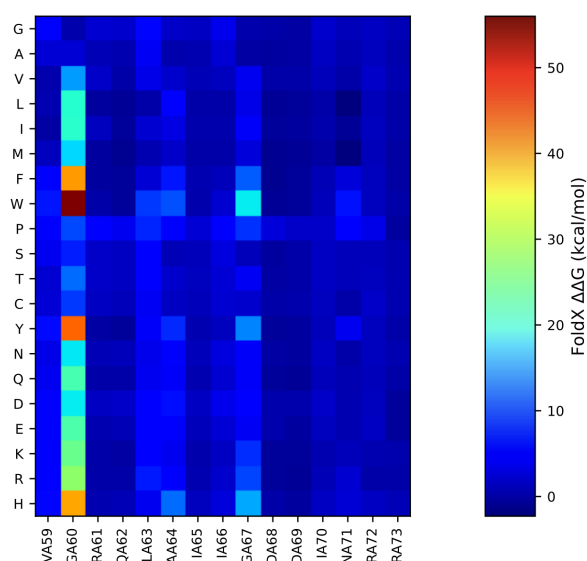
To assess whether the predicted effects of BH3 mutations depend on the conformational state of BAK1, MutateX calculations were extended to an ensemble-based framework. The analysis used the eight cluster structures obtained from the molecular dynamics simulations described in Section 3.6. For each structure, saturation mutagenesis was performed on the BH3 motif, and the resulting  $\Delta\Delta G$  values were averaged across the ensemble. This approach considers multiple conformations instead of a single static structure, resulting in a more reliable analysis of mutation effects. Because the ensemble was generated from the 2IMS crystal structure, the residue numbering differs from the UniProt numbering used in previous sections. In the ensemble models, the analyzed BH3 segment corresponds to residues 59–73, with the sequence VGRQLAIGDDINRR.

#### 3.6.1. Ensemble-averaged $\Delta\Delta G$ profile of the BH3 motif

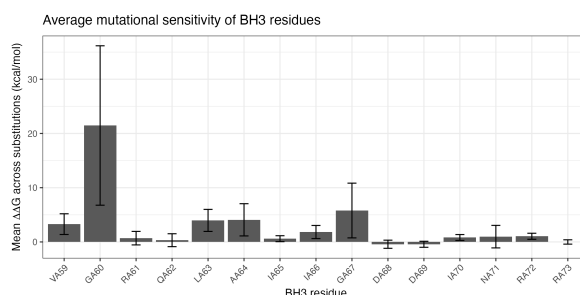
The ensemble-mode heatmap (Figure 8) shows that the BH3 motif is not uniformly sensitive to mutation. Most positions display low to moderate  $\Delta\Delta G$  values for many substitutions, indicating that several side-chain changes can be accommodated without strongly disrupting the local structure. This is consistent with the clustering analysis, where the ensemble mainly consisted of structurally similar conformations.

However, clear mutation-sensitive positions are also observed. GA60 and GA67 show the strongest destabilizing effects, especially when replaced by bulky aromatic residues such as tryptophan or tyrosine. This suggests that the small size of glycine is structurally important at these sites. Introducing larger side chains is likely to create steric clashes and disturb the local packing of the BH3 helix.

To quantify the mutational sensitivity of individual BH3 positions more directly, the average  $\Delta\Delta G$  value across all substitutions was calculated for each residue (Figure 9). GA60 and GA67 display the highest average destabilization, confirming their role as the principal structural hotspots within the BH3 motif. In contrast, residues with lower average  $\Delta\Delta G$  values appear more tolerant to substitution, suggesting greater local structural permissiveness. The variability observed between substitutions tells us that mutation effects depends on the chemical properties of the introduced amino acid. A general trend observed across the ensemble analysis is the strong destabilizing effect associated with proline substitutions. At nearly all analyzed positions,



**Figure 8.** Ensemble-mode MutateX heatmap. Warmer colours indicate more destabilizing mutations, whereas darker blue regions correspond to substitutions with limited energetic impact.

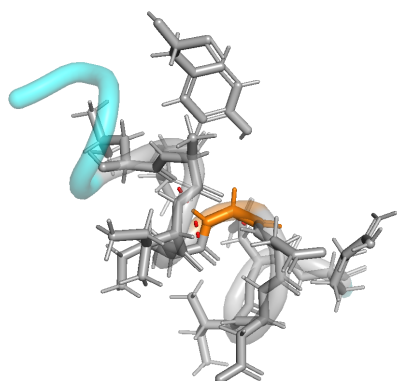


**Figure 9.** Mean  $\Delta\Delta G$  values per BH3 residue across all amino acid substitutions. Error bars represent the standard deviation between substitutions.

replacement by proline produced higher  $\Delta\Delta G$  values, which is because proline is structurally unusual compared with other amino acids. In addition, bulky aromatic substitutions, such as tryptophan, tyrosine and phenylalanine, shows increased destabilization at glycine positions such as GA60 and GA67. This suggests that the small size of glycine is important for preserving the local structure of the BH3 helix. To better understand the structural effects of the mutations observed in the ensemble analysis, Pymol was used to visualize one of the hotspots (Figure 10). Substitution of GA67 by tryptophan introduces the bulky aromatic side chain into a structurally constrained region of the BH3 helix. The larger side chain likely disturbs nearby residues and affects the local structure of the  $\alpha$ -helix. These observations provide a structural explanation for the strongly positive  $\Delta\Delta G$  values associated with GA67W in both the single-structure and ensemble-mode analyses.

#### 3.6.2. Comparison with the single-structure stability analysis

Comparison of the ensemble-mode analysis with the single-structure MutateX results presented in 3.3 reveals a strong overall agreement in the mutational sensitivity profile of the BH3 motif. In both analyses,



**Figure 10.** The substitution of glycine 67 by tryptophan as visualized in Pymol. Red dashed lines indicate close contacts between the introduced tryptophan residue and neighboring amino acids.

glycine residues such as GA60 and GA67 appeared to be the most sensitive to mutation, showing consistently high  $\Delta\Delta G$  values across many substitutions. This suggests that the small size of glycine is important for maintaining the local structure of the BH3 helix. However, the ensemble-mode calculations provide a more detailed representation of mutation effects by taking into account the structural changes observed during the molecular dynamics simulations. Mutations that remain more unstable across the ensemble are more likely to represent structurally important hotspot positions that remain consistent across multiple conformations. In this context, GA60 and GA67 show the most consistently sensitive behavior within the BH3 motif. The ensemble analysis also suggests that certain mutations may have a smaller effect because of local structural flexibility. Although the most destabilizing mutations were still observed in the ensemble analysis, several moderate mutations showed lower  $\Delta\Delta G$  values compared with the single-structure analysis. This suggests that small structural changes within the ensemble may help some side-chain substitutions.

## 4. Discussion

This study combined pathogenicity prediction, structure selection, docking, saturation mutagenesis, and ensemble-based analysis to investigate the structural constraints acting on BAK1. The BH motifs consistently emerged as the most sensitive regions, supporting their essential role in BAK1 activation and dimerisation. The agreement between AlphaMissense and DeMaSk further shows how these specific motifs are under strong structural and evolutionary constraint.

### 4.1. Structure Selection and Methodological Framework

In order to perform the downstream analysis and generate reliable results, structure selection was the first step. 2IMT was selected as the main free-protein structure because it had the best overall quality and coverage, while 6UXR was selected to represent the activated dimer for the interaction analysis. 2IMS was additionally included for the ensemble-generation workflow, as it provided a suitable starting conformation for conformational sampling.

The molecular dynamics simulations and subsequent clustering analysis confirm that the monomeric BAK1 structure is highly stable and remains unfolded during the simulation. Indeed, this unfolding characteristic is shown by the consistently low fluctuations in both the root

mean square deviation and the radius of gyration, coupled with the preservation of core residue contacts. Furthermore, the overwhelming dominance of a single conformational state, occupying over 85% of the simulated trajectory, suggests that wild-type BAK1 resides in a deep, well-defined free energy minimum. The rare, transient excursions into minor clusters likely represent the natural, localized motions characteristic of folded proteins. Establishing this baseline of global rigidity, with some minor dynamic fluctuations, was crucial, as it provided a robust and realistic structural ensemble for accurately evaluating the energetic impacts of specific mutations.

### 4.2. Variant-Specific Effects and State-Dependent Pathogenicity

The BH3 motif plays a central role in BAK1 activation and dimerization through its interaction with the hydrophobic groove of neighbouring BAK1 monomers. Mutations that destabilize this region are therefore expected to affect the ability of BAK1 to adopt or maintain conformations required for oligomerization and mitochondrial outer membrane permeabilization.

A particularly informative case is the variant G126A in the BH1, reported in cBioPortal and classified as pathogenic in the literature that can be found in Mavisp [12]. In our analysis, this substitution is predicted as neutral for stability (Section 3.3), consistent with the broad tolerance of GA126 to substitution due to its solvent-exposed position. However, the same variant shows a destabilising effect on the binding interface (Section 3.4). This decoupling suggests that G126A does not compromise the fold of monomeric BAK1 but rather impairs its ability to engage in protein–protein contacts. We therefore hypothesise that the pathogenicity of G126A arises from disruption of dimer formation, which is essential for BAK1-mediated pore assembly and apoptosis execution; the loss of dimerisation would prevent downstream mitochondrial outer membrane permeabilisation despite the protein remaining structurally intact. In contrast, the V74M variant, consistently identified across all three clinical databases, shows the opposite pattern. Our results indicate that V74M is destabilizing for the protein's overall fold but remains neutral for the binding interface. This suggests its pathogenicity is likely due to structural instability or protein degradation rather than a direct failure to interact. Even if the binding site remains functional, the lack of a stable protein pool would ultimately prevent BAK1 from executing its pro-apoptotic role.

A similar rationale extends to other reported variants in our dataset: substitutions at exposed positions tend to be tolerated for fold stability while still potentially compromising function through interface effects, whereas bulky side chains introduced at buried positions are predicted to be severely destabilising and incompatible with a folded, functional protein. A strength of our analysis is that it was performed on both the inactive monomeric and the activated oligomeric forms of BAK1, providing a more complete view of mutational effects across its conformational cycle than a single-structure analysis would allow. Crucially, residues that appear solvent-exposed and tolerant in the monomeric form may become buried or engaged in critical contacts upon activation, meaning that neutrality in the resting state is not a reliable indicator of functional tolerance. This state-dependent behaviour explains the pathogenicity of several variants in our dataset, as well as existing ones such as G126A, where neutral monomer stability masks disruption of the activated dimer interface, and underscores the value of integrating multiple conformational states when interpreting missense variants in dynamic proteins like BAK1.

### 4.3. Mutational Scanning and Structural Hotspots in the BH3 Region

The ensemble-mode analysis indicated that the strongest constraints were concentrated at a limited number of positions, particularly GA60 and GA67, which appear essential for preserving the geometry and packing of the BH3 helix. In contrast, several other positions within

the BH3 motif were comparatively tolerant to substitution, suggesting localized flexibility within the region. MutateX further demonstrated that this tolerance was not uniform across the BH motifs, as glycine and hydrophobic residues were especially important for maintaining structural integrity, whereas proline substitutions were broadly destabilizing due to their incompatibility with  $\alpha$ -helical geometry.

Interface analysis similarly revealed that binding stability depends strongly on residue identity and packing interactions. While certain positions were highly sensitive to mutations, others accommodated variation more readily, indicating a balance between structural rigidity and conformational flexibility at the dimer interface. Collectively, these analyses identify the BH3 region as the primary structural hotspot within BAK1. The strong agreement among pathogenicity prediction, mutational scanning, docking, and ensemble analyses supports the conclusion that a small subset of residues is under substantial functional and evolutionary constraint.

## 5. Conclusion

This study identified the BH motifs, and particularly the BH3 and BH1 region, as key determinants of BAK1 stability and activation. By combining pathogenicity prediction, structural analysis, docking, saturation mutagenesis, and ensemble-based approaches, residues were identified as highly mutation-sensitive and structurally constrained. Specifically, MutateX analysis highlighted major hotspots such as LA78 and WA125, where mutations severely compromise both monomer stability and dimer interaction, marking them as essential structural anchors. A central contribution of this work is the use of both the inactive monomeric and the activated oligomeric forms of BAK1, which proved essential for capturing variants whose effects only emerge upon conformational change and dimerisation. Applying this dual-state framework, we were able to rationalise and catalogue reported pathogenic variants, including cases where neutrality in the resting fold masks a clear disruption of the activated interface. Overall, the results provide insight into the structural mechanisms underlying BAK1 activation, highlight specific residues critical for preserving its apoptotic function, and demonstrate that integrating multiple conformational states is a powerful strategy for interpreting missense variants in conformationally dynamic proteins.

## 6. Author Contributions:

- **Emma Jinschek (s253599)** - Structure Selection, Discussion on Structure Selection and Methodological Framework
- **André Godinho (s253707)** - Prediction of pathogenicity, Stability using MutateX, Discussion on Variant-Specific Effects and State-Dependent Pathogenicity
- **Maxence Marbouty (s253730)** - Variant selection, Local interaction using MutateX, Discussion on Variant-Specific Effects and State-Dependent Pathogenicity
- **Louis Page (s253729)** - Ensemble Generation, Discussion on Structure Selection and Methodological Framework (Molecular dynamics part)
- **Veronica Mescu (s215159)** - Ensemble-mode mutational stability analysis, Discussion on Mutational Scanning and Structural Hotspots in the BH3 Region
- **Abstract, Introduction and Conclusion:** All the coauthors contributed equally \*

## 7. Data Availability

The source code used to generate the results reported in this study is publicly available on GitHub: [https://github.com/andregodinhodtu/22117\\_protein\\_structure\\_group7](https://github.com/andregodinhodtu/22117_protein_structure_group7).

Additionally, project files are stored on the DTU Healthtech server at: `/home/projects/22117_protein_structure/projects/group07/22117_protein_structure_group7` on `pupil2.healthtech.dtu`.

## 8. Declaration of AI Usage

AI tools (ChatGPT and Claude) were used to assist with clarifying ideas, improving grammar, and refining the flow of the text. All scientific content, analyses, and interpretations are the work of the authors.

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## A. Supplementary Material

### A.1. Consensus Clinical Variant List

The following table lists the priority missense variants identified through the consensus filtering approach. They were retrieved from ClinVar, COSMIC, and cBioPortal, and selected based on their presence in at least two of these databases.

| Variant | Database of Origin          |
|---------|-----------------------------|
| V74M    | ClinVar, COSMIC, cBioPortal |
| R76W    | ClinVar, COSMIC, cBioPortal |
| V142I   | ClinVar, COSMIC, cBioPortal |
| E120V   | ClinVar, COSMIC             |
| G135S   | ClinVar, COSMIC             |
| G82R    | ClinVar, COSMIC             |
| H145R   | ClinVar, COSMIC             |
| N124I   | ClinVar, COSMIC             |
| Q144R   | ClinVar, COSMIC             |
| N124S   | ClinVar, cBioPortal         |
| R127P   | ClinVar, cBioPortal         |
| R87L    | ClinVar, cBioPortal         |
| A139T   | COSMIC, cBioPortal          |
| A79T    | COSMIC, cBioPortal          |
| D84N    | COSMIC, cBioPortal          |
| G122D   | COSMIC, cBioPortal          |
| G82E    | COSMIC, cBioPortal          |
| G82W    | COSMIC, cBioPortal          |
| I81T    | COSMIC, cBioPortal          |
| L78F    | COSMIC, cBioPortal          |
| N86D    | COSMIC, cBioPortal          |
| R137C   | COSMIC, cBioPortal          |
| R137S   | COSMIC, cBioPortal          |
| R87Q    | COSMIC, cBioPortal          |
| R88H    | COSMIC, cBioPortal          |
| S69I    | COSMIC, cBioPortal          |
| W125L   | COSMIC, cBioPortal          |

**Table 1.** Final set of 27 clinical variants selected as a priority subset for MutateX analysis.